

C++ Developer Test

At Lumu Technologies our goal is to illuminate threats and adversaries. To do so, being able to look at what is happening inside a client machine is paramount.

Show us your strength by writing a command line program that monitors the HTTP activity in the machine for a specified time and prints out a summary.

You are free to enrich the output summary as much as you see fit but it must include at least:

- The total number of HTTP requests detected
- An histogram of the top 10 requested hosts, by number of queries, descending, e.g.

```
domain1.com 110  *****
domain2.com 82   *****
domain3.com 51   *****
....
```

Suggested command syntax:

```
<program name> [/L:<log file path>] <monitor time in seconds>
```

Remarks

Hint: to monitor network traffic you can use the PcapPlusPlus library (<https://pcapplusplus.github.io/>).

You can build it for your preferred platform and invest as much time as you'd like. Please log your progress using commits in any CVS.

You can use third party software to help you in your task, but you must log any such usage. You can of course use the Internet as much as you like.

Your deliverables are:

- A running copy of your program so we can test what you built
- A way for us to review your code through a control version system (for example, Github) where we can review your incremental progress
- A minimal documentation on how to build and use it, and its limitations, if any
- The references to third-party software you are using, if any
- An evaluation of the complexity of the algorithms you are using to generate the histograms
- Anything else you'd like to share with us

Please also be prepared to show your application running and discuss your code with us in some depth.

Questions? Need help? Let us know! We're here to help.